

Полное решение

Модуль 1 + Модуль 2 + Модуль 3

Сетевое и системное администрирование · ALT Linux

ISP / HQ-RTR / BR-RTR	ALT JeOS 10.4
HQ-SRV / BR-SRV	ALT Server 10.4
HQ-CLI	ALT Workstation 10.4
Домен	au-team.irpo
Пароль (везде)	P@ssw0rd
SSH порт	2026

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1 Общие настройки — все машины

```

Hostname (подставить своё имя)

hostnamectl set-hostname hq-rtr.au-team.irpo

# Имена: isp hq-rtr br-rtr hq-srv br-srv hq-cli (+ .au-team.irpo)

Часовой пояс

timedatectl set-timezone Asia/Yekaterinburg

```

2 ISP — NAT в интернет

```

ISP — интерфейсы (etcnet)

# /etc/net/sysctl.conf → net.ipv4.ip_forward = 1

# ens18: TYPE=eth BOOTPROTO=dhcp ONBOOT=yes

# ens19: TYPE=eth BOOTPROTO=static ONBOOT=yes

echo '172.16.1.1/28' > /etc/net/ifaces/ens19/ipv4address

# ens20: TYPE=eth BOOTPROTO=static ONBOOT=yes

echo '172.16.2.1/28' > /etc/net/ifaces/ens20/ipv4address

systemctl restart network

ISP — NAT

apt-get install -y iptables

iptables -t nat -A POSTROUTING -o ens18 -j MASQUERADE

iptables-save > /etc/sysconfig/iptables && systemctl enable iptables

```

3 HQ-RTR — WAN + VLAN 100/200/999

```

HQ-RTR — сеть

# sysctl.conf → net.ipv4.ip_forward = 1

# ens18: TYPE=eth BOOTPROTO=static ONBOOT=yes

echo '172.16.1.2/28' > /etc/net/ifaces/ens18/ipv4address

echo 'default via 172.16.1.1' > /etc/net/ifaces/ens18/ipv4route

echo 'nameserver 77.88.8.7' > /etc/net/ifaces/ens18/resolv.conf

# ens19: TYPE=eth BOOTPROTO=static ONBOOT=yes (trunk)

# ens19.100: TYPE=vlan HOST=ens19 VID=100 BOOTPROTO=static ONBOOT=yes

echo '192.168.100.1/27' > /etc/net/ifaces/ens19.100/ipv4address

# ens19.200: TYPE=vlan HOST=ens19 VID=200 BOOTPROTO=static ONBOOT=yes

echo '192.168.200.1/27' > /etc/net/ifaces/ens19.200/ipv4address

# ens19.999: TYPE=vlan HOST=ens19 VID=999 BOOTPROTO=static ONBOOT=yes

echo '192.168.99.1/29' > /etc/net/ifaces/ens19.999/ipv4address

iptables -t nat -A POSTROUTING -o ens18 -j MASQUERADE

iptables-save > /etc/sysconfig/iptables && systemctl enable iptables

systemctl restart network && ip -br a

```

4 BR-RTR — WAN + LAN

BR-RTR — сеть

```
# sysctl.conf → net.ipv4.ip_forward = 1
# ens18: TYPE=eth BOOTPROTO=static ONBOOT=yes
echo '172.16.2.2/28' > /etc/net/ifaces/ens18/ipv4address
echo 'default via 172.16.2.1' > /etc/net/ifaces/ens18/ipv4route
# ens19: TYPE=eth BOOTPROTO=static ONBOOT=yes
echo '192.168.30.1/28' > /etc/net/ifaces/ens19/ipv4address
iptables -t nat -A POSTROUTING -o ens18 -j MASQUERADE
iptables-save > /etc/sysconfig/iptables && systemctl enable iptables
systemctl restart network && ip -br a
```

5-7 HQ-SRV / BR-SRV / HQ-CLI — сеть

HQ-SRV — VLAN100

```
# ens18: TYPE=eth BOOTPROTO=static ONBOOT=yes
# ens18.100: TYPE=vlan HOST=ens18 VID=100 BOOTPROTO=static ONBOOT=yes
echo '192.168.100.10/27' > /etc/net/ifaces/ens18.100/ipv4address
echo 'default via 192.168.100.1' > /etc/net/ifaces/ens18.100/ipv4route
systemctl restart network
```

BR-SRV

```
# ens18: TYPE=eth BOOTPROTO=static ONBOOT=yes
echo '192.168.30.10/28' > /etc/net/ifaces/ens18/ipv4address
echo 'default via 192.168.30.1' > /etc/net/ifaces/ens18/ipv4route
systemctl restart network
```

HQ-CLI — VLAN200 DHCP

```
# ens18: TYPE=eth BOOTPROTO=static ONBOOT=yes
# ens18.200: TYPE=vlan HOST=ens18 VID=200 BOOTPROTO=dhcp ONBOOT=yes
systemctl restart network && ip -br a
```

8 SSH + пользователи

sshuser (HQ-SRV и BR-SRV)

```
useradd sshuser -u 2026
echo 'sshuser:P@ssw0rd' | chpasswd
echo 'sshuser ALL=(ALL:ALL) NOPASSWD: ALL' > /etc/sudoers.d/sshuser
```

SSH hardening — /etc/openssh/sshd_config

```
# Port 2026
# AllowUsers sshuser
# MaxAuthTries 2
# Banner /etc/openssh/banner
echo 'Authorized access only' > /etc/openssh/banner
sshd -t && systemctl restart sshd
```

net_admin (HQ-RTR и BR-RTR)

```
useradd net_admin && echo 'net_admin:P@ssw0rd' | chpasswd
echo 'net_admin ALL=(ALL:ALL) NOPASSWD: ALL' > /etc/sudoers.d/net_admin
```

10 GRE туннель HQ ↔ BR

```
HQ-RTR - /etc/net/ifaces/gre1/
# options: TYPE=iptun TUNTYPE=gre TUNLOCAL=172.16.1.2 TUNREMOTE=172.16.2.2 TUNOPTIONS='ttl 64'
echo '10.255.255.1/30' > /etc/net/ifaces/gre1/ipv4address
systemctl restart network

BR-RTR - /etc/net/ifaces/gre1/
# options: TYPE=iptun TUNTYPE=gre TUNLOCAL=172.16.2.2 TUNREMOTE=172.16.1.2 TUNOPTIONS='ttl 64'
echo '10.255.255.2/30' > /etc/net/ifaces/gre1/ipv4address
systemctl restart network && ping -c 2 10.255.255.1
```

11

OSPF (FRR) + MD5

```
06a поурча - FRR
apt-get install -y frr
# /etc/frr/daemons → ospfd=yes
systemctl enable --now frr && systemctl restart frr
```

```
HQ-RTR - vtysh
vtysh
conf t
interface ens18
ip ospf passive
interface ens19
ip ospf passive
router ospf
ospf router-id 1.1.1.1
network 10.255.255.0/30 area 0
network 192.168.100.0/27 area 0
network 192.168.200.0/27 area 0
network 192.168.99.0/29 area 0
interface gre1
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 P@ssw0rd
ip ospf network point-to-point
end
```

```
write memory
```

```
BR-RTR - vtysh
vtysh
conf t
interface ens18
ip ospf passive
interface ens19
ip ospf passive
router ospf
ospf router-id 2.2.2.2
network 10.255.255.0/30 area 0
network 192.168.30.0/28 area 0
interface gre1
ip ospf authentication message-digest
```

```
ip ospf message-digest-key 1 md5 P@ssw0rd
ip ospf network point-to-point
end
write memory
```

Проверка OSPF

```
vtysh -c 'show ip ospf neighbor'
ip r | grep ospf
```

13 DHCP VLAN200 (HQ-RTR)

```
HQ-RTR ~ /etc/dhcp/dhcpd.conf
apt-get install -y dhcp-server
subnet 192.168.200.0 netmask 255.255.255.224 {
    range 192.168.200.50 192.168.200.100;
    option routers 192.168.200.1;
    option domain-name "au-team.irpo";
    option domain-name-servers 192.168.100.10;
}
# /etc/sysconfig/dhcpd → DHCPD_INTERFACE=ens19.200
dhcpd -t && systemctl enable --now dhcpd
```

14 DNS dnsmasq (HQ-SRV)

```
HQ-SRV ~ /etc/dnsmasq.conf.d/au-team.conf
apt-get install -y dnsmasq bind-utils
domain=au-team.irpo
local=/au-team.irpo/
bind-interfaces
listen-address=127.0.0.1,192.168.100.10
no-resolv
server=77.88.8.7
server=77.88.8.3
address=/hq-rtr.au-team.irpo/192.168.100.1
address=/br-rtr.au-team.irpo/192.168.30.1
address=/hq-srv.au-team.irpo/192.168.100.10
address=/hq-cli.au-team.irpo/192.168.200.10
address=/br-srv.au-team.irpo/192.168.30.10
address=/docker.au-team.irpo/172.16.1.1
address=/web.au-team.irpo/172.16.2.1
ptr-record=1.100.168.192.in-addr.arpa,hq-rtr.au-team.irpo
ptr-record=10.100.168.192.in-addr.arpa,hq-srv.au-team.irpo
ptr-record=10.200.168.192.in-addr.arpa,hq-cli.au-team.irpo
cache-size=1000
dnsmasq --test && systemctl enable --now dnsmasq
```

M2.1 Samba DC + пользователи + sudo (BR-SRV → HQ-CLI)

i BR-SRV: ALT Server 10.4. Пакет: task-samba-dc. НЕ запускать smbd/winbind/nmbd при роли DC.

BR-SRV — провизия домена

```
apt-get install -y task-samba-dc
rm -f /etc/samba/smb.conf && rm -rf /var/lib/samba/* /var/cache/samba/*
samba-tool domain provision
# Realm: AU-TEAM.IRPO | Domain: AU-TEAM | Role: dc
# DNS: SAMBA_INTERNAL | Forwarder: 77.88.8.7 | Password: P@ssw0rd
cp /var/lib/samba/private/krb5.conf /etc/krb5.conf
systemctl enable --now samba
echo 'search au-team.irpo' > /etc/net/ifaces/ens18/resolv.conf
echo 'nameserver 127.0.0.1' >> /etc/net/ifaces/ens18/resolv.conf
systemctl restart network
```

BR-SRV — группа hq + 5 пользователей

```
samba-tool group add hq
for i in {1..5}; do
samba-tool user add hquser$i P@ssw0rd
samba-tool user setexpiry hquser$i --noexpiry
samba-tool group addmembers hq hquser$i
done
```

HQ-RTR — DNS в DHCP на BR-SRV

```
sed -i 's/domain-name-servers.*/domain-name-servers 192.168.30.10;/' /etc/dhcp/dhcpd.conf
systemctl restart dhcpd
```

HQ-CLI — ввод в домен

```
apt-get install -y task-auth-ad-sssd
# ЦУС → Пользователи → Аутентификация → Домен AD → au-team.irpo → SSSD → Настроить
# Перезагрузить HQ-CLI
roleadd hq wheel
echo 'Cmnd_Alias SHELLCMD = /bin/cat, /bin/grep, /usr/bin/id' > /etc/sudoers.d/hq
echo 'WHEEL_USERS ALL=(ALL) SHELLCMD' >> /etc/sudoers.d/hq
```

M2.2 RAID 0 → /raid (HQ-SRV)

HQ-SRV — mdadm

```
apt-get install -y mdadm && lsblk
mdadm --create /dev/md0 --level=0 --raid-devices=2 /dev/sdb /dev/sdc
mdadm --detail --scan | tee -a /etc/mdadm.conf
mkfs.ext4 /dev/md0 && mkdir /raid
echo '/dev/md0 /raid ext4 defaults 0 0' >> /etc/fstab
mount -av && df -h | grep raid
```

M2.3 NFS /raid/nfs (HQ-SRV → HQ-CLI)

HQ-SRV – сервер

```
apt-get install -y nfs-server
mkdir -p /raid/nfs && chmod -R 777 /raid/nfs
echo '/raid/nfs 192.168.200.0/24(rw,no_root_squash)' > /etc/exports
exportfs -arv && systemctl enable --now nfs-server
```

HQ-CLI – автомонтирование

```
mkdir -p /mnt/nfs
echo '192.168.100.10:/raid/nfs /mnt/nfs nfs defaults,_netdev 0 0' >> /etc/fstab
mount -av && df -h | grep nfs
```

M2.4 NTP chrony стратум 5 (ISP сервер → все клиенты)

ISP – сервер

```
apt-get install -y chrony
sed -i 's/^pool/#pool/' /etc/chrony.conf
cat <<EOF >> /etc/chrony.conf
server ntp0.ntp-servers.net iburst prefer minstratum 4
local stratum 5
allow 0.0.0.0/0
EOF
systemctl restart chronyd && chronyc tracking
```

Клиенты (HQ-SRV HQ-CLI BR-RTR BR-SRV)

```
apt-get install -y chrony
sed -i 's/^pool/#pool/' /etc/chrony.conf
# HQ-узлы → 172.16.1.1 | BR-узлы → 172.16.2.1
echo 'server 172.16.1.1 iburst' >> /etc/chrony.conf
systemctl restart chronyd && chronyc sources
```

M2.5 Ansible (BR-SRV)

BR-SRV – ansible.cfg + hosts

```
apt-get install -y ansible sshpass
cat <<EOF > /etc/ansible/ansible.cfg
[defaults]
inventory = /etc/ansible/hosts
host_key_checking = False
[all:vars]
ansible_python_interpreter=/usr/bin/python3
EOF
cat <<EOF > /etc/ansible/hosts
HQ-SRV ansible_host=192.168.100.10 ansible_user=sshuser ansible_password=P@ssw0rd ansible_port=2026
HQ-CLI ansible_host=192.168.200.10 ansible_user=sshuser ansible_password=P@ssw0rd ansible_port=2026
HQ-RTR ansible_host=192.168.100.1 ansible_user=net_admin ansible_password=P@ssw0rd
BR-RTR ansible_host=192.168.30.1 ansible_user=net_admin ansible_password=P@ssw0rd
EOF
```

```
cd /etc/ansible && ansible -m ping all
```

M2.6 Docker testapp + db (BR-SRV)

```
BR-SRV — docker + импорт образов
apt-get install -y docker-engine docker-compose
systemctl enable --now docker
mount /dev/cdrom /mnt/cdrom
docker load < /mnt/cdrom/docker/site_latest.tar
docker load < /mnt/cdrom/docker/mariadb_latest.tar
BR-SRV — /opt/testapp/compose.yaml
mkdir -p /opt/testapp && cat <<EOF > /opt/testapp/compose.yaml
services:
  database:
    container_name: db
    image: mariadb:10.11
    restart: always
    environment:
      MARIADB_DATABASE: testdb
      MARIADB_USER: testc
      MARIADB_PASSWORD: P@ssw0rd
      MARIADB_ROOT_PASSWORD: toor
  app:
    container_name: testapp
    image: site:latest
    restart: always
    ports:
      - '8080:8080'
    environment:
      DB_TYPE: maria
      DB_HOST: 192.168.30.10
      DB_PORT: '3306'
      DB_NAME: testdb
      DB_USER: testc
      DB_PASS: P@ssw0rd
    depends_on: [database]
EOF
cd /opt/testapp && docker compose up -d && docker compose ps
```

M2.7 Apache + MariaDB веб-приложение (HQ-SRV)

△ ALT Server 10.4: apache=httpd, mysql=mysqlld. Корень: /var/www/html. Пакет: lamp-server (всё сразу)

```
HQ-SRV — lamp-server + webdb
apt-get install -y lamp-server
systemctl enable --now httpd mysqld
mount /dev/cdrom /mnt/cdrom
```

```
mysql -u root <<EOF
CREATE DATABASE webdb;
CREATE USER 'web'@'localhost' IDENTIFIED BY 'P@ssw0rd';
GRANT ALL ON webdb.* TO 'web'@'localhost'; FLUSH PRIVILEGES;
EOF
mysql -u root webdb < /mnt/cdrom/web/dump.sql
cp /mnt/cdrom/web/index.php /var/www/html/
cp -r /mnt/cdrom/web/images /var/www/html/
# Исправить в index.php: user='web' password='P@ssw0rd' dbname='webdb'
sed -i "s/\$username = .*/\$username = 'web';/" /var/www/html/index.php
sed -i "s/\$password = .*/\$password = 'P@ssw0rd';/" /var/www/html/index.php
sed -i "s/\$dbname = .*/\$dbname = 'webdb';/" /var/www/html/index.php
systemctl restart httpd
```

M2.8 Port forwarding (HQ-RTR и BR-RTR)

```
HQ-RTR — DNAT
iptables -t nat -A PREROUTING -i ens18 -p tcp --dport 8080 -j DNAT --to 192.168.100.10:80
iptables -t nat -A PREROUTING -i ens18 -p tcp --dport 2026 -j DNAT --to 192.168.100.10:2026
iptables -A FORWARD -d 192.168.100.10 -j ACCEPT
iptables-save > /etc/sysconfig/iptables

BR-RTR — DNAT
iptables -t nat -A PREROUTING -i ens18 -p tcp --dport 8080 -j DNAT --to 192.168.30.10:8080
iptables -t nat -A PREROUTING -i ens18 -p tcp --dport 2026 -j DNAT --to 192.168.30.10:2026
iptables -A FORWARD -d 192.168.30.10 -j ACCEPT
iptables-save > /etc/sysconfig/iptables
```

M2.9 Nginx reverse proxy (ISP)

```
ISP — /etc/nginx/conf.d/proxy.conf
apt-get install -y nginx
cat <<EOF > /etc/nginx/conf.d/proxy.conf
server {
listen 80; server_name web.au-team.irpo;
location / { proxy_pass http://192.168.100.10; }
}
server {
listen 80; server_name docker.au-team.irpo;
location / { proxy_pass http://192.168.30.10:8080; }
}
EOF
nginx -t && systemctl enable --now nginx
```

M2.10 Web-аутентификация htpasswd (ISP)

```
ISP — htpasswd + auth_basic
apt-get install -y apache2-htpasswd
```

```
htpasswd -bc /etc/nginx/.htpasswd WEB P@ssw0rd
# Добавить в секцию server web.au-team.irpo:
auth_basic "Restricted";
auth_basic_user_file /etc/nginx/.htpasswd;
nginx -t && systemctl reload nginx
curl -I http://web.au-team.irpo # 401
curl -u WEB:P@ssw0rd http://web.au-team.irpo # 200
```

M2.11 Яндекс Браузер (HQ-CLI)

```
HQ-CLI
mount /dev/cdrom /mnt/cdrom
find /mnt/cdrom -name 'yandex*.rpm'
rpm -i /mnt/cdrom/yandex-browser*.rpm
# Или: apt-get install -y yandex-browser-stable
```

МЗ.3

Шифрованный туннель IPsec поверх GRE (HQ-RTR ↔ BR-RTR)

! Самый простой путь: IPsec в режиме transport поверх существующего GRE. OSPF не трогать.

Оба роутера – strongswan

```
apt-get install -y strongswan
```

HQ-RTR – /etc/ipsec.conf

```
config setup
```

```
conn gre-protect
```

```
left=172.16.1.2
```

```
right=172.16.2.2
```

```
authby=secret
```

```
type=transport
```

```
auto=start
```

HQ-RTR – /etc/ipsec.secrets

```
172.16.1.2 172.16.2.2 : PSK "P@ssw0rd"
```

BR-RTR – /etc/ipsec.conf

```
config setup
```

```
conn gre-protect
```

```
left=172.16.2.2
```

```
right=172.16.1.2
```

```
authby=secret
```

```
type=transport
```

```
auto=start
```

BR-RTR – /etc/ipsec.secrets

```
172.16.2.2 172.16.1.2 : PSK "P@ssw0rd"
```

Запуск и проверка

```
systemctl enable --now strongswan
```

```
ipsec status # оба роутера
```

```
ipsec statusall | grep ESTABLISHED
```

МЗ.4

Межсетевой экран iptables (HQ-RTR и BR-RTR)

⚠ Применять ПОСЛЕ настройки туннеля и port forwarding. Иначе потеряешь доступ.

HQ-RTR – минимальный firewall

```
iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

```
iptables -A INPUT -i lo -j ACCEPT
```

```
iptables -A INPUT -p icmp -j ACCEPT
```

```
iptables -A INPUT -p tcp -m multiport --dports 2026,80,443,53 -j ACCEPT
```

```
iptables -A INPUT -p udp -m multiport --dports 53,123 -j ACCEPT
```

```
iptables -A FORWARD -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
```

```
iptables -A FORWARD -i ens19.100 -o ens18 -j ACCEPT
```

```

iptables -A FORWARD -i ens19.200 -o ens18 -j ACCEPT
iptables -A FORWARD -i ens18 -m conntrack --ctstate NEW -j DROP
iptables -P INPUT DROP
iptables-save > /etc/sysconfig/iptables && systemctl enable iptables

BR-RTR — аналогично (ens19 без VLAN)

iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -p icmp -j ACCEPT
iptables -A INPUT -p tcp -m multiport --dports 2026,80,443,53 -j ACCEPT
iptables -A INPUT -p udp -m multiport --dports 53,123 -j ACCEPT
iptables -A FORWARD -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
iptables -A FORWARD -i ens19 -o ens18 -j ACCEPT
iptables -A FORWARD -i ens18 -m conntrack --ctstate NEW -j DROP
iptables -P INPUT DROP
iptables-save > /etc/sysconfig/iptables && systemctl enable iptables

```

М3.5 CUPS + PDF-принтер (HQ-SRV → HQ-CLI)

```

HQ-SRV — установка CUPS

apt-get install -y cups cups-pdf
# /etc/cups/cupsd.conf → Listen *:631, Allow all в Location /
systemctl enable --now cups

lpadmin -p PDF -E -v cups-pdf:/ -m CUPS-PDF.ppd
lpadmin -p PDF -o printer-is-shared=true
lpstat -p

HQ-CLI — принтер по умолчанию

apt-get install -y cups-client

lpadmin -p PDF -E -v ipp://192.168.100.10:631/printers/PDF -m everywhere
lpoptions -d PDF && lpstat -d

```

М3.6 rsyslog + logrotate (HQ-SRV сервер)

```

HQ-SRV — сервер

apt-get install -y rsyslog
sed -i '/imudp/s/^#//' /etc/rsyslog.conf
sed -i '/imtcp/s/^#//' /etc/rsyslog.conf
printf '$template R,"/opt/%sHOSTNAME%/%sPROGRAMNAME%.log"\n*.warn ?R\n& stop\n' >
/etc/rsyslog.d/remote.conf
mkdir -p /opt && systemctl restart rsyslog
ss -lnup | grep 514

/etc/logrotate.d/opt-logs

/opt/*/*.log {
weekly
compress
missingok
minsize 10M
}

```

Клиенты — HQ-RTR BR-RTR BR-SRV

```
apt-get install -y rsyslog
echo '*.warn @@192.168.100.10:514' > /etc/rsyslog.d/client.conf
systemctl restart rsyslog
logger -p warning 'test' && sleep 2 && ls /opt/
```

М3.7 Мониторинг Netdata (HQ-SRV)

HQ-SRV — netdata + nginx + DNS

```
apt-get install -y netdata nginx
# /etc/netdata/netdata.conf → bind to = 0.0.0.0
systemctl enable --now netdata
# Добавить DNS запись в dnsmasq
echo 'address=mon.au-team.irpo/192.168.100.10' >> /etc/dnsmasq.conf.d/au-team.conf
systemctl restart dnsmasq
cat <<EOF > /etc/nginx/conf.d/mon.conf
server {
listen 80; server_name mon.au-team.irpo;
auth_basic "Monitoring";
auth_basic_user_file /etc/nginx/.htpasswd-mon;
location / { proxy_pass http://127.0.0.1:19999; }
}
EOF
htpasswd -bc /etc/nginx/.htpasswd-mon admin P@ssw0rd
nginx -t && systemctl enable --now nginx && systemctl reload nginx
```

М3.8 Ansible PC-INFO инвентаризация (BR-SRV)

⚠ Требуется М2.5: рабочий Ansible на BR-SRV

BR-SRV — из iso или вручную

```
mount /dev/cdrom /mnt/cdrom
cp /mnt/cdrom/playbook/*.yaml /etc/ansible/ 2>/dev/null || true
mkdir -p /etc/ansible/PC-INFO
# Если плейбука нет в iso — создать:
cat <<'EOF' > /etc/ansible/pc-info.yml
---
- hosts: HQ-SRV,HQ-CLI
gather_facts: yes
tasks:
- copy:
content: "hostname: {{ ansible_hostname }}\nip: {{ ansible_default_ipv4.address }}\n"
dest: /etc/ansible/PC-INFO/{{ ansible_hostname }}.yaml
delegate_to: localhost
EOF
cd /etc/ansible && ansible-playbook pc-info.yml
ls PC-INFO/ && cat PC-INFO/hq-srv.yaml
```

М3·9 fail2ban — защита SSH (HQ-SRV)

```
HQ-SRV — 1 файл
apt-get install -y fail2ban
cat <<EOF > /etc/fail2ban/jail.d/ssh.conf
[sshd]
enabled = true
port = 2026
maxretry = 3
bantime = 60
backend = systemd
EOF
systemctl enable --now fail2ban
fail2ban-client status sshd
```

М3·10 Кибер Бэкап 17.4 (HQ-SRV + HQ-CLI)

△ Требуется М2·7: рабочий Apache + MariaDB + база webdb. Выполнять последним.

```
Алгоритм по памяти
mount /dev/cdrom /mnt/cdrom
rpm -i /mnt/cdrom/CyberBackup-ManagementServer*.rpm # HQ-SRV
rpm -i /mnt/cdrom/CyberBackup-Agent*.rpm # HQ-CLI
# Браузер HQ-CLI → https://192.168.100.10:9877
# 1. Создать организацию: irpo
# 2. Создать юзера: irpoadmin / P@ssw0rd / Administrator
# 3. Добавить устройство HQ-CLI
mkdir -p /backup && chmod 777 /backup # на HQ-CLI
# 4. Хранилища → Добавить → /backup на HQ-CLI
# 5. План 1: Файлы → /etc → /backup → Запустить
# 6. План 2: Базы данных → MySQL → webdb → /backup → Запустить
```

Финальный чеклист

Устройство	Команда	Результат
M1	<code>ip -br a && ip r</code>	адреса на всех устройствах
M1	<code>ping 10.255.255.2 с HQ-RTR</code>	GRE туннель UP
M1	<code>vtysh -c 'show ip ospf neighbor'</code>	OSPF FULL
M1	<code>ip r grep ospf</code>	маршруты в таблице
M1	<code>ping 192.168.30.10 с HQ-SRV</code>	HQ → BR доступность
M1	<code>cat /etc/resolv.conf на HQ-CLI</code>	nameserver 192.168.100.10
M1	<code>ss -lntp grep 2026 на HQ-SRV</code>	SSH слушает
M1	<code>dig @192.168.100.10 ya.ru +short</code>	DNS работает
M2	<code>samba-tool domain info 127.0.0.1</code>	домен поднят
M2	<code>id hquser1@au-team.irpo на HQ-CLI</code>	вход в домен
M2	<code>mdadm --detail /dev/md0</code>	RAID0 State: clean
M2	<code>df -h grep /mnt/nfs</code>	NFS смонтирован

Устройство	Команда	Результат
M2	<code>chronyc tracking</code> на ISP	Stratum: 5
M2	<code>ansible -m ping all</code> с BR-SRV	все pong
M2	<code>docker compose ps</code> на BR-SRV	testapp + db Up
M2	<code>curl http://192.168.100.10</code>	сайт Apache
M2	<code>curl -I http://web.au-team.irpo</code>	nginx прокси
M3	<code>ipsec status</code>	ESTABLISHED
M3	<code>fail2ban-client status sshd</code>	jail активен
M3	<code>systemctl status netdata</code>	мониторинг UP
M3	<code>lpstat -d</code> на HQ-CLI	PDF принтер default
M3	<code>ls /opt/</code> на HQ-SRV	папки логов